UrbanZero

Learning City Driving from Scratch with Reinforcement Learning and ROS2

Summary

This project trains an autonomous vehicle agent to navigate urban environments entirely through reinforcement learning — no expert demonstrations, no rulebooks, no hardcoded behaviors. Using the CARLA simulator with a ROS2 bridge, a PPO agent learns to drive through a city (intersections, traffic, traffic lights, pedestrians) from scratch, guided only by a reward signal. The project compares naive vs. shaped reward formulations and evaluates how well emergent driving behavior generalizes across different city maps.

Motivation

Current autonomous driving systems rely heavily on hand-engineered rules, imitation learning from human drivers, or massive labeled datasets — all of which are expensive, brittle, and hard to generalize. A tabula rasa RL approach asks a more fundamental question: can safe, functional driving behavior emerge purely from interaction with an environment? Beyond academic interest, this has real implications for AV development in edge cases where human demonstration data is sparse or unavailable — unusual intersections, novel road layouts, emergent traffic scenarios.

Understanding how and when RL policies succeed or fail in urban driving directly informs the next generation of autonomous systems. As cities grow more complex and vehicles more autonomous, methods that do not require human supervision become increasingly valuable.

Challenge

City driving is one of the hardest RL problems because the state space is enormous (other vehicles, pedestrians, signals, road geometry), the reward signal is sparse and delayed, and bad policies can cause cascading failures (one wrong turn leads to being permanently stuck). Designing a reward function that produces safe, traffic-aware behavior without hand-engineering every rule is non-trivial — naive rewards lead to agents that game the metric (e.g., stopping permanently to avoid collision penalties). Bridging the gap between simulation fidelity and meaningful learned behavior adds further complexity.

Expected Approach

CARLA will serve as the primary simulator, using Town01 for training and Town02/Town03 for generalization testing. The CARLA-ROS2 bridge will expose sensor data (camera, LiDAR, collision and lane sensors) as ROS2 topics, which feed into a custom Gymnasium environment wrapper. A PPO agent via Stable-Baselines3 will be trained with two reward configurations: (1) a naive reward based solely on forward progress and collision penalty, and (2) a shaped reward adding terms for lane adherence, traffic light compliance, and proximity to other agents.

Dynamic traffic will be spawned using CARLA's Traffic Manager. Training curves, policy rollouts, and generalization metrics (success rate, infraction count) will be logged via TensorBoard and compared across both reward schemes.

Related Work

Dosovitskiy et al. introduced CARLA as a benchmark for autonomous driving research, establishing it as the standard platform for learning-based AV experiments. The simulator supports flexible sensor suites, dynamic traffic, and multiple urban map configurations, making it ideal for RL training. Reinforcement learning for urban driving has been explored in depth by Toromanoff, Wirbel, and Moutarde (CVPR 2020), who demonstrated that model-free RL agents using implicit affordances can achieve competitive results on the CARLA leaderboard without expert supervision — to their knowledge the first successful RL agent handling traffic light detection and pedestrian avoidance simultaneously.

The broader landscape of deep RL for autonomous driving is surveyed comprehensively by Kiran et al. (IEEE TITS 2022), who provide a taxonomy of DRL approaches across behavior planning, motion planning, and control tasks, and highlight the key computational challenges in real-world deployment. The PPO algorithm itself (Schulman et al., 2017) has become the standard policy gradient method for continuous control tasks due to its stability and ease of implementation relative to prior methods like TRPO.

Milestones / Timeline

Milestone	Start Date	End Date	Description
CARLA + ROS2 Setup	Feb 27	Mar 6	CARLA running locally, ROS2 bridge functional, sensor topics verified, ego vehicle spawning confirmed
Gym Wrapper	Mar 6	Mar 13	Custom Gymnasium env: obs from camera+LiDAR, actions as throttle/steer/brake, reset logic and episode termination working
Naive Reward Baseline	Mar 13	Mar 20	PPO training with distance-progress + collision penalty only; agent successfully completes basic straight road navigation
Dynamic Traffic Integration	Mar 20	Mar 27	CARLA Traffic Manager spawning vehicles and pedestrians; baseline policy evaluated against dynamic obstacles
Shaped Reward Training	Mar 27	Apr 10	Train shaped reward variant (lane keeping, traffic light, proximity terms); compare learning curves to baseline
Generalization Eval	Apr 10	Apr 17	Test both policies on Town02 and Town03 (unseen maps); record success rate, collisions, and infractions
Final Demo + Writeup	Apr 17	Apr 24	Demo video of city driving policy, final written report, polished GitHub README with all contributions documented

Materials Needed

CARLA Simulator (0.9.15), CARLA-ROS2 Bridge

ROS2 Humble

Python, Stable-Baselines3, Gymnasium

TensorBoard (training visualization)

NVIDIA RTX 4080 (local training)

Git / GitHub (version control)

References

[1] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An Open Urban Driving Simulator," in Proc. 1st Annual Conference on Robot Learning (CoRL), vol. 78, pp. 1–16, 2017.

[2] M. Toromanoff, E. Wirbel, and F. Moutarde, "End-to-End Model-Free Reinforcement Learning for Urban Driving Using Implicit Affordances," in Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7153–7162, 2020.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," arXiv:1707.06347, 2017.

[4] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. A. Sallab, S. Yogamani, and P. Perez, "Deep Reinforcement Learning for Autonomous Driving: A Survey," IEEE Trans. Intelligent Transportation Systems, vol. 23, no. 6, pp. 4909–4926, 2022.